



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Building Multi Agent Applications with Gemini ADK (C829)

Course Code: C829

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	11 August 2026	Initial release aligned across slide deck, Learner Guide, Lesson Plan, and eight connected labs.	Tertiary Infotech Trainer

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Course Information

Course Title	Building Multi Agent Applications with Gemini ADK (C829)
Course Reference	C829
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 7.5 scheduled hours per day (15 hours)
Daily Timing	9:30 am – 6:00 pm (1-hour lunch; tea breaks within scheduled time)
Mode	Instructor-led, hands-on practical labs
Trainer	Tertiary Infotech Trainer

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Explain the ADK agent lifecycle and configure a Gemini agent with clear instructions, typed input or output, and session-aware behaviour.
- LO2: Build reliable function tools and API integrations with explicit schemas, validation, error handling, and safe secret management.
- LO3: Use session state and searchable memory to preserve relevant context without confusing conversation history, working state, and long-term knowledge.
- LO4: Design multi-agent systems with specialist roles, coordinator delegation, shared context, and deterministic graph workflows where control matters.
- LO5: Connect ADK agents to MCP capabilities and choose an appropriate routing strategy for tools, specialists, and workflow branches.
- LO6: Apply evaluation, observability, security guardrails, and deployment practices to prepare a multi-agent application for production use.

Learning Reinforcement

- Every topic is followed by hands-on labs that learners complete on their own machine.
- Each lab ends with a 'Test it' verification step so learners can confirm their own progress.
- The trainer circulates during lab time to give individual feedback and unblock learners.
- Each topic closes with a recap mapped back to the course learning outcomes.

Course Schedule

Day 1 — Single Agents, Tools, State and Memory

Time	Duration	Topic / Activity
9:30–9:50	20 min	Welcome, environment check, course roadmap and learning outcomes
9:50–10:50	60 min	Topic 1 — Single Agent Foundations (concepts, architecture and worked example)
10:50–11:05	15 min	Tea break
11:05–12:05	60 min	Hands-on: Lab 1: Create the SupportOps Agent
12:05–13:00	55 min	Hands-on: Lab 2: Add Structured Triage and Sessions
13:00–14:00	60 min	Lunch break
14:00–15:00	60 min	Topic 2 — Tools, Memory and Sessions (concepts, patterns and worked example)
15:00–15:15	15 min	Tea break
15:15–16:20	65 min	Hands-on: Lab 3: Connect Reliable Function Tools
16:20–17:40	80 min	Hands-on: Lab 4: Add Searchable Knowledge and Memory
17:40–18:00	20 min	Day 1 recap, troubleshooting clinic and reflection

Total scheduled time: 450 minutes (7.5 hours) excluding lunch — of which 420 minutes (7 hours) are instructional, plus 30 minutes of tea breaks.

Day 2 — Multi-Agent Orchestration, MCP and Production Readiness

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap, checkpoint recovery and Day 2 orientation
9:45–10:45	60 min	Topic 3 — Multi-Agent Architecture (concepts, pattern selection and worked example)
10:45–11:00	15 min	Tea break
11:00–12:05	65 min	Hands-on: Lab 5: Build a Specialist Agent Team
12:05–13:00	55 min	Hands-on: Lab 6: Orchestrate a Deterministic Resolution Pipeline
13:00–14:00	60 min	Lunch break
14:00–15:00	60 min	Topic 4 — MCP and Production Design (concepts, production trade-offs and worked example)
15:00–15:15	15 min	Tea break
15:15–16:20	65 min	Hands-on: Lab 7: Discover Tools Through MCP
16:20–17:40	80 min	Hands-on: Lab 8: Harden, Observe, and Package SupportOps
17:40–18:00	20 min	Course recap, production checklist, action plan and Q&A

Total scheduled time: 450 minutes (7.5 hours) excluding lunch — of which 420 minutes (7 hours) are instructional, plus 30 minutes of tea breaks.

Lab Reference (aligned to course topics)

Topic	Coverage	Labs
Topic 01: Single Agent Foundations	2 of 8 labs (25%)	Lab 1, Lab 2
Topic 02: Tools, Memory and Sessions	2 of 8 labs (25%)	Lab 3, Lab 4
Topic 03: Multi-Agent Architecture	2 of 8 labs (25%)	Lab 5, Lab 6
Topic 04: MCP and Production Design	2 of 8 labs (25%)	Lab 7, Lab 8